

# Technical Documentation

Diesel Engine  
**12 V 4000 R20**

Operating Instructions  
**M015380/00E**



Printed in Germany

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Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen.

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Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

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Imprimé en Allemagne

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Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

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El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

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Stampato in Germania

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

## Notes for the user

This Manual consists of 11 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents.

## Using the Table of Contents to trace an entry

- ➡ Table of Contents (general)
- ➡ Section
- ➡ Contents
- ➡ Heading

## Using the Alphabetical Index to trace an entry

- ➡ Search term
- ➡ Section
- ➡ Heading

## Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

## ! Safety Instructions

Summary

## A Product Summary

Engine summary, technical data

## C Operating Instructions

Task descriptions

## D Maintenance Schedule

Time schedule and maintenance task chart

## E Troubleshooting

Causes, remedies

## F Drawings

Engine cutaway drawings

## G Task Description

Inspection and maintenance tasks

## K Preservation

Engine preservation

## T Transportation Locking Device

Drawing, instructions

## ? Keyword Index

Alphabetical index of key terms

## Z Appendix

Installation drawings, diagrams



# Wichtig – Important – Importante

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**Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen GmbH zurücksenden.**

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

**Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.**

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

**Veillez séparer la carte “Signalisation de mise en service” et la renvoyer à la MTU Friedrichshafen GmbH.**

Les informations concernant la signalisation de mise en service constituent la base pour l'assistance en exploitation contractuelle (garantie, rechanges, etc.).

**Rogamos separen la tarjeta “Aviso de puesta en servicio” y la devuelvan rellena a MTU Friedrichshafen GmbH.**

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

**Staccare “Avviso di messa in servizio” e rispedirlo debitamente compilato alla MTU Friedrichshafen GmbH.**

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).



|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| <div></div> | <div></div>                                                                                              |
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*Scrivere in stampatello!*



|                                                                                                                                            |                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Motornr.:<br>Engine No.:<br>N° du moteur:<br>N° de motor:<br>Motore N.:                                                                    | Auftragsnr.:<br>MTU works order No.:<br>N° de commande:<br>N° de pedido:<br>N. commessa:                                                                           |
| Motortyp:<br>Engine model:<br>Type du moteur:<br>Tipo de motor:<br>Motore tipo:                                                            | Inbetriebnahmedatum:<br>Date put into operation:<br>Mise en service le:<br>Fecha de puesta en servicio:<br>Messa in servizio il:                                   |
| Eingebaut in:<br>Installation site:<br>Lieu de montage:<br>Lugar de montaje:<br>Installato:                                                | Schiffstyp / Schiffshersteller:<br>Vessel/type/class / Shipyard:<br>Type du bateau / Constructeur:<br>Tipo de bugue / Constructor:<br>Tipo di Barca / Costruttore: |
| Endabnehmer/Anschrift:<br>End user's address:<br>Adresse du client final:<br>Dirección del cliente final:<br>Indirizzo del cliente finale: |                                                                                                                                                                    |
| Bemerkung:<br>Remarks:<br>Remarques:<br>Observaciones:<br>Commento:                                                                        |                                                                                                                                                                    |

**Inbetriebnahmemeldung**

**Commissioning Note**

**Notice de mise en service**

**Aviso de puesta en servicio**

**Avviso di messa in servizio**

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**These Safety Instructions must be read and followed by any persons operating or carrying out maintenance on the machinery plant.**

## Basic Safety Precautions

This MTU engine is a state-of-the-art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of

- Incorrect use
- Operation by unqualified personnel
- Modifications or conversions
- Noncompliance with the Safety Instructions

## Correct Use

The engine is intended solely for the use specified in the contract. Any other use is considered improper use. The manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Correct use also includes observation of the operating instructions and compliance with the maintenance instructions.

## Personnel Requirements

Work on the engine must be carried out only by reliable personnel. The specified legal minimum age must be respected.

Only fully-trained or qualified personnel must be employed. Responsibilities of the operating, maintenance and repair personnel must be specified.

## Modifications or Conversions

Modifications made by the customer to the engine may affect safety. No modifications or conversions must be implemented out without prior consent from MTU.

No liability will be accepted for damage resulting from unauthorised modifications.

## Organisational Measures

The Operation Manual must be made available to the operating personnel. It must be kept at hand near the engine and accessible to the operating and maintenance personnel at all times.

The personnel must be instructed on engine operation by means of the Operation Manual, and in particular the safety instructions must be explained. This is especially important for personnel who work on the engine only on an occasional basis.

In addition to the Operation Manual, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

## Auxiliary Materials

Fluids and lubricants must be kept in properly designated containers. When using fluids, lubricants and other chemical substances, follow the safety instructions applicable to the product. Take special care when using hot and caustic materials. When using inflammable materials, avoid all sparks and do not smoke.

Use only fluids and lubricants that have been tested and approved by MTU and are listed in the Fluids and Lubricants Specification A001061.

## **Spare Parts**

Spare parts must at least satisfy the requirements specified by the manufacturer. This is guaranteed when original components are used.

## **Working Clothes**

Work clothing must be tight-fitting so that it does not catch on rotating or projecting components. Jewelry must not be worn.

## **Transportation**

Lift the engine only with the lifting eyes provided. Use only the transport and lifting equipment approved by MTU. The engine must only be transported in installation position.

## **Engine Operation**

When the engine is running, ear protectors must always be worn in the engine room.

The engine room must be well ventilated.

To avoid injuries caused by falling, leaked or spilt engine oil and coolant must be cleaned up immediately or absorbed with appropriate bonding agents.

Exhaust gases from combustion engines are poisonous and injurious to health if inhaled. The exhaust pipework must be leak-free and must discharge the gases to atmosphere.

During engine operation, do not touch battery terminals, generator terminals or cables. Inadequate protection of electrical components can lead to electric shocks and serious injuries.

## **Maintenance**

Compliance with maintenance specifications is an important safety factor.

Unless expressly permitted, maintenance work must not be carried out with the engine running. The engine must be secured against inadvertent starting and the battery disconnected. Persons not involved must keep clear.

Never attempt to rectify faults or carry out repairs if you do not have the necessary experience or special tools required. Maintenance work must be carried out by authorised, qualified personnel. Use only tools in perfect condition.

Fluids emerging under high pressure can penetrate clothing and skin and may cause serious injury. Before starting work, relieve pressure in systems and H.P. lines which are to be opened.

Until the engine has cooled, the breather cap must not be opened. Release the breather cap seal and allow pressure to escape, before cap removal.

Take special care when draining hot fluids – risk of injury. Drain the fluids into a suitable container and wipe up any spillages.

When changing engine oil or working on the fuel system, ensure that the engine room is adequately ventilated.

When working with compressed air, always wear protective goggles. The pressure must not exceed 3 bar. When working with liquid nitrogen, always wear protective clothing – risk of injury.

When working high on the engine, always use suitable ladders and work platforms. Make sure components are placed on stable surfaces.

Gases released from the battery are explosive. Avoid sparks and naked flames. Do not allow battery acids to come into contact with skin or clothing. Wear protective goggles. Do not place tools on the battery. Before connecting the cable to the battery, check battery polarity. Battery pole reversal may lead to injury through the sudden discharge of acid or bursting of the battery body.

Before barring the engine, make sure that nobody is standing in the danger zone. After working on the engine, check that all guards have been reinstalled and that all tools and loose components have been removed from the engine.